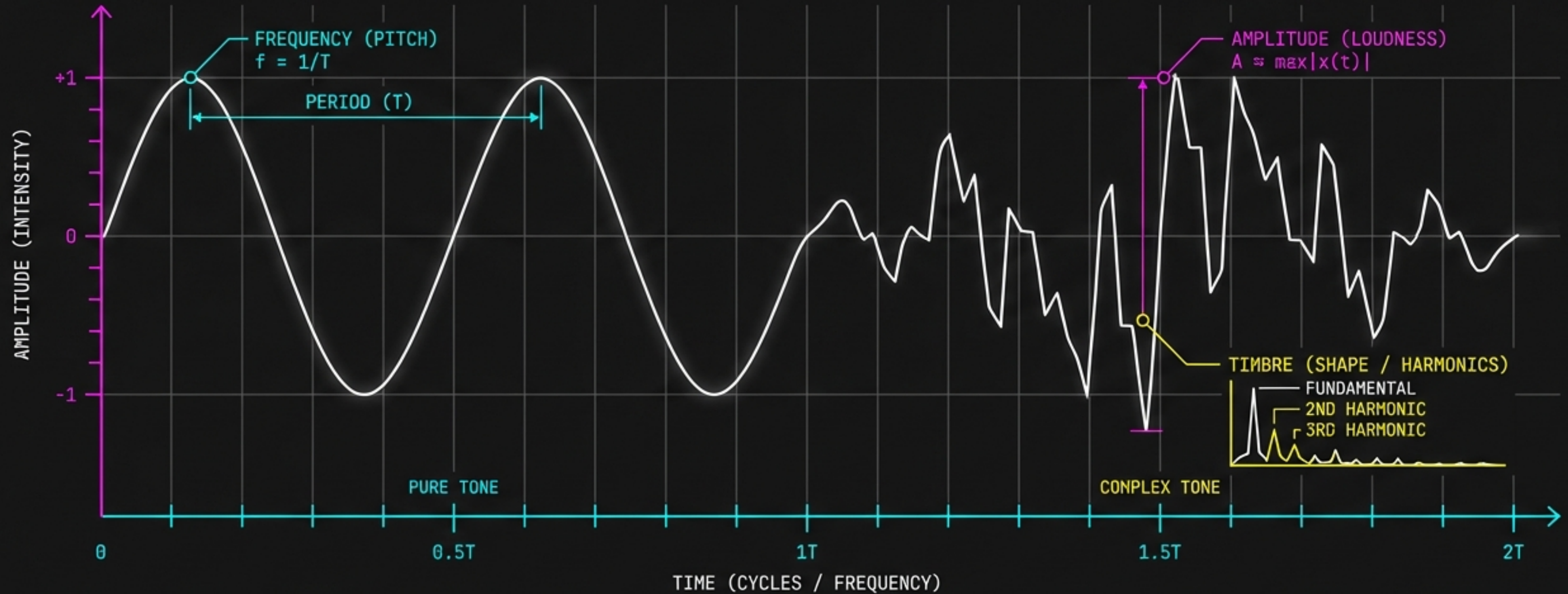


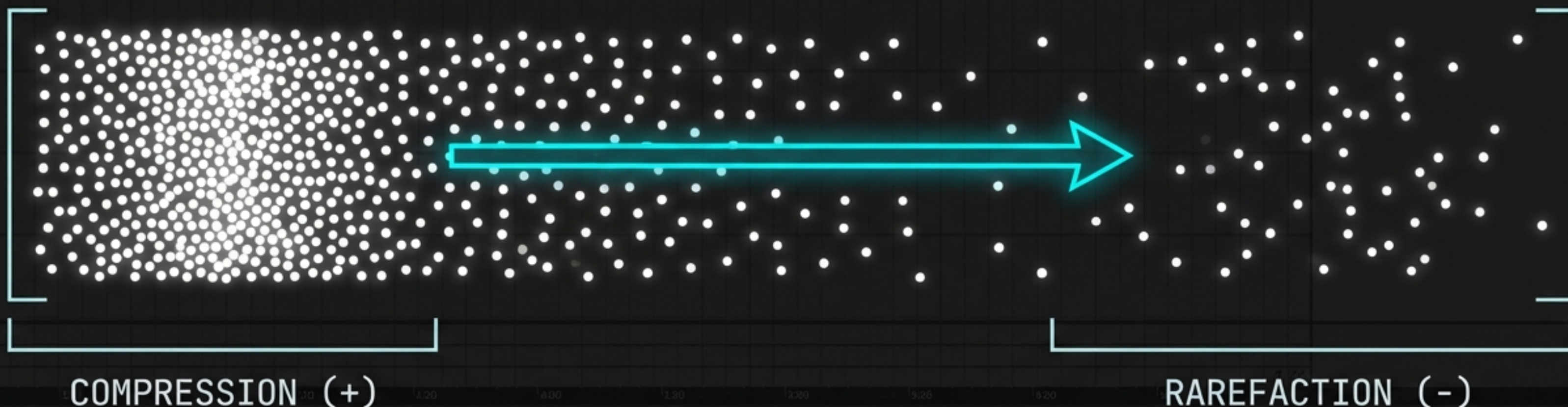
SOUND WAVE FUNDAMENTALS

The Physics of Electronic Music



DECONSTRUCTING FREQUENCY // AMPLITUDE // TIMBRE

THE INVISIBLE MEDIUM

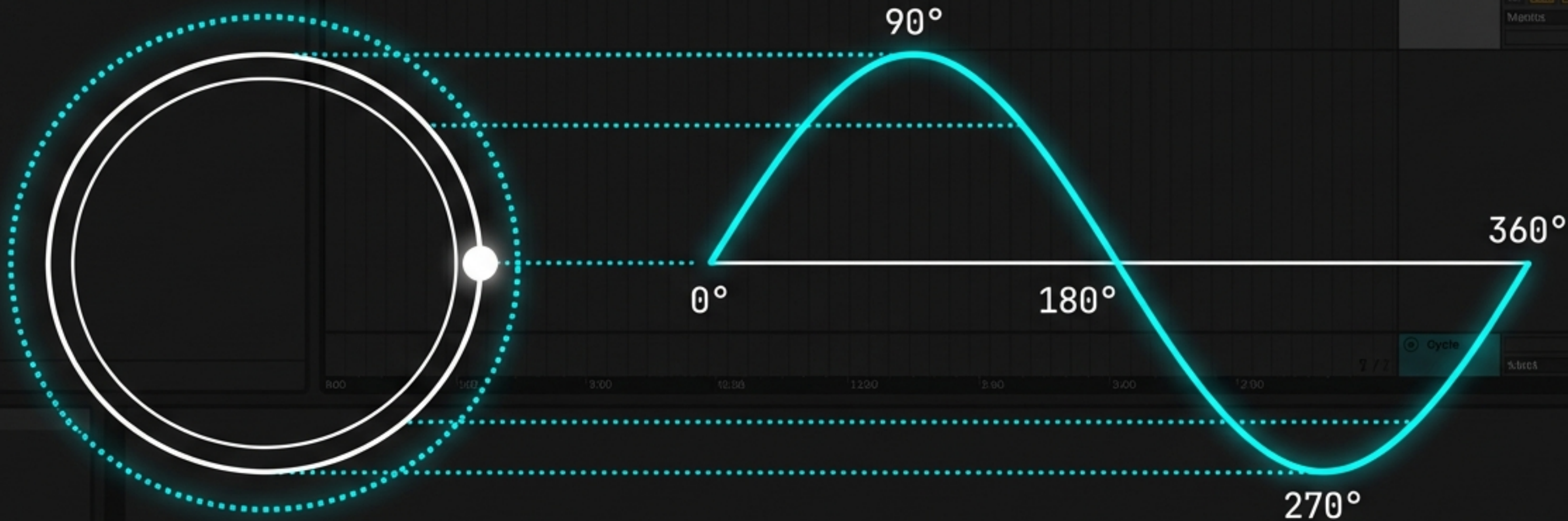


**SOUND IS MECHANICAL
PERTURBATION.**

NO AIR = NO SOUND.

THE CYCLE

THE ATOM OF SOUND



A Cycle = ONE complete oscillation from start, through all phases, back to start.

TWO SIDES OF THE SAME COIN

FREQUENCY (Hz)

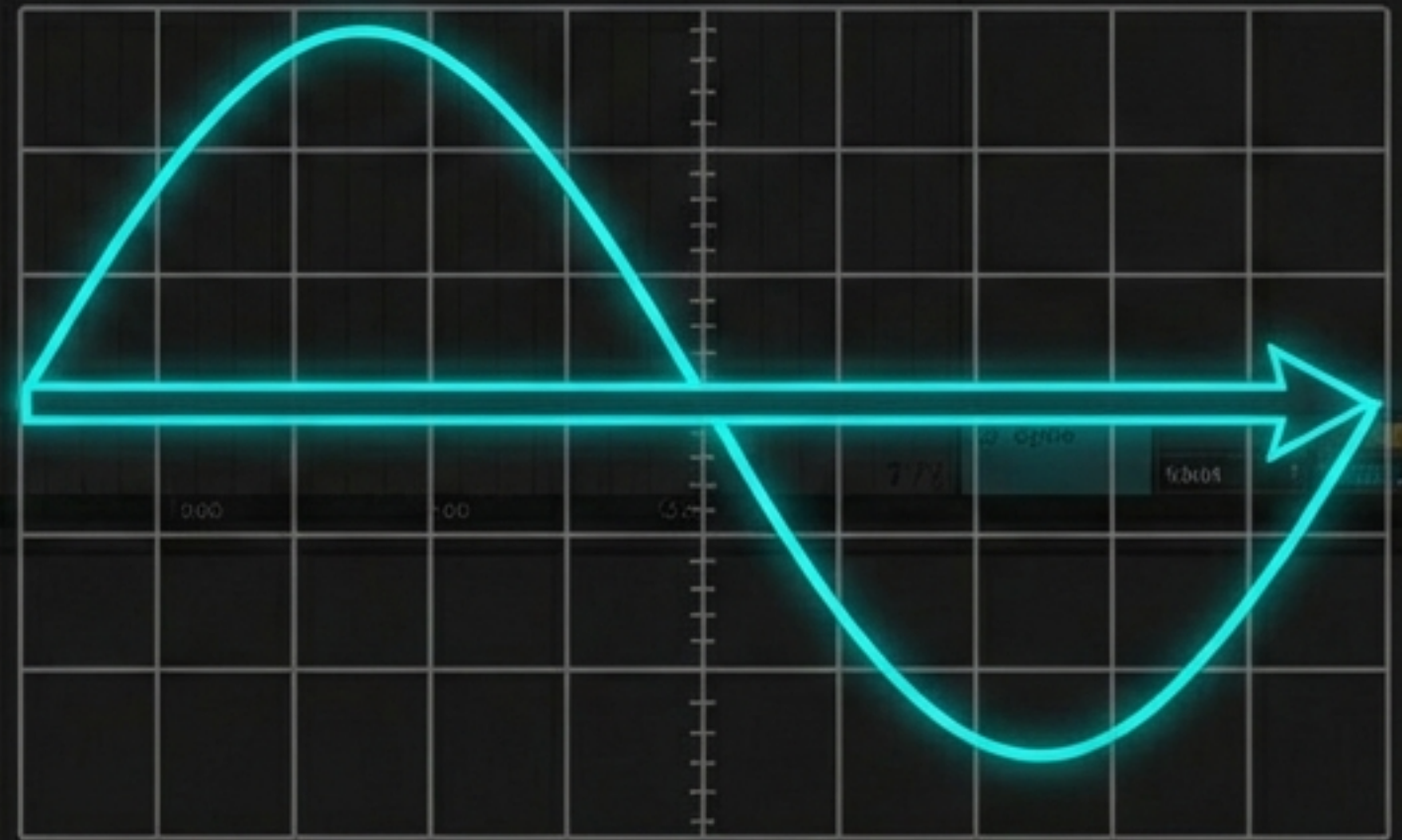
How Many?



Engine Speed (RPM)

PERIOD (s)

How Long?



Time for one rotation

THE RECIPROCAL RULE

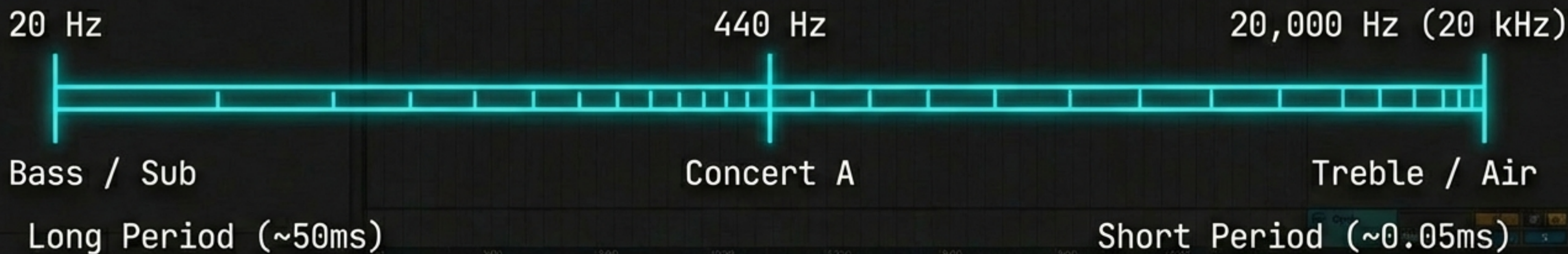
$$f = 1 / T$$

$$T = 1 / f$$

$$100 \text{ Hz} = 0.01 \text{ s (10 ms)}$$

Product is always 1.

THE CANVAS: HUMAN HEARING RANGE



Higher Frequency = Higher Pitch

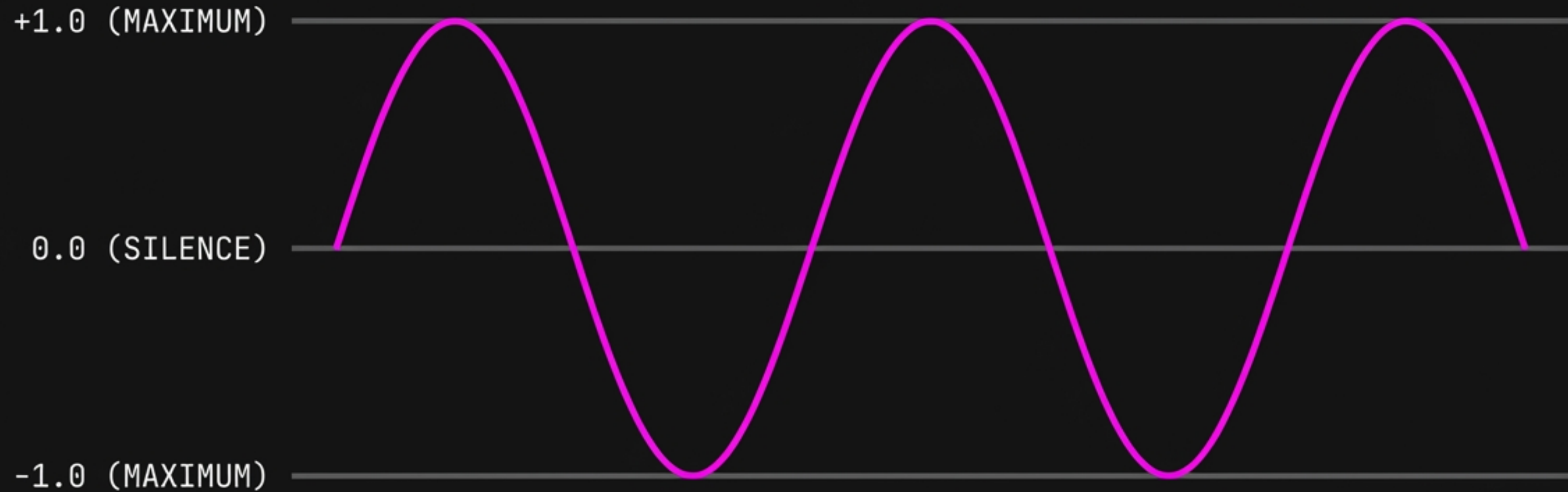
Dimension 2: Intensity



Amplitude measures pressure variation.

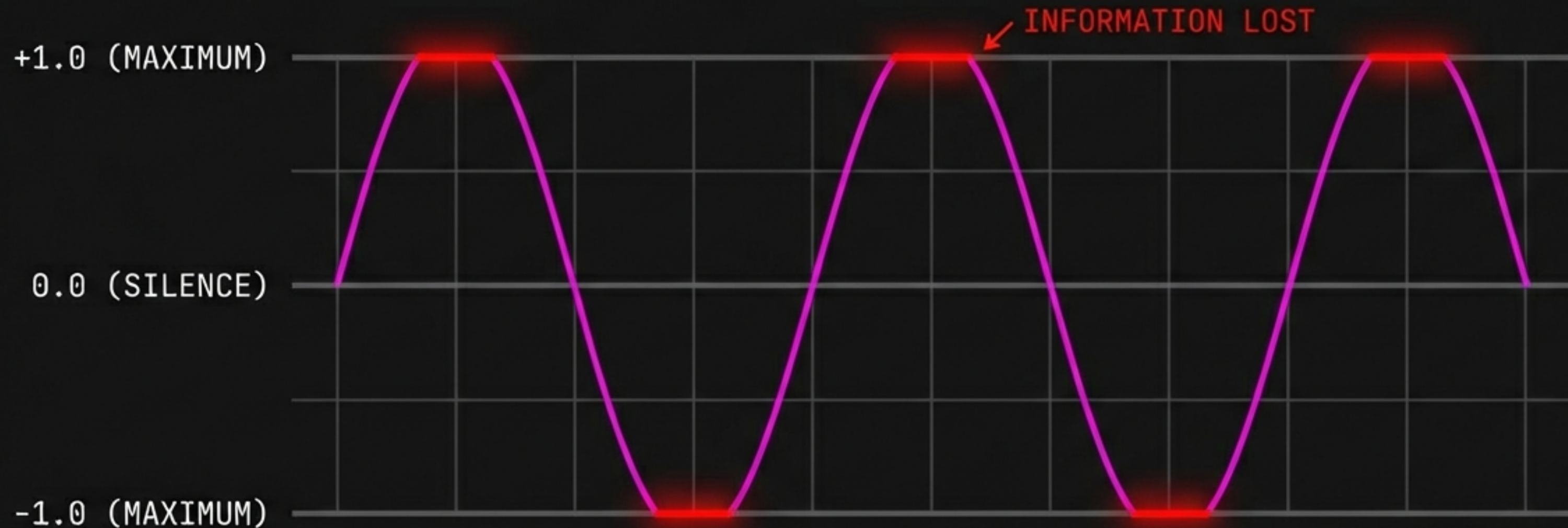
Vertical Height = Loudness

THE DIGITAL BOX



Digital audio is a decimal game between -1 and +1.

The Danger Zone: Clipping



Waveform: DEFORMED
Sound: DISTORTED

Thinking in Decibels (dB)



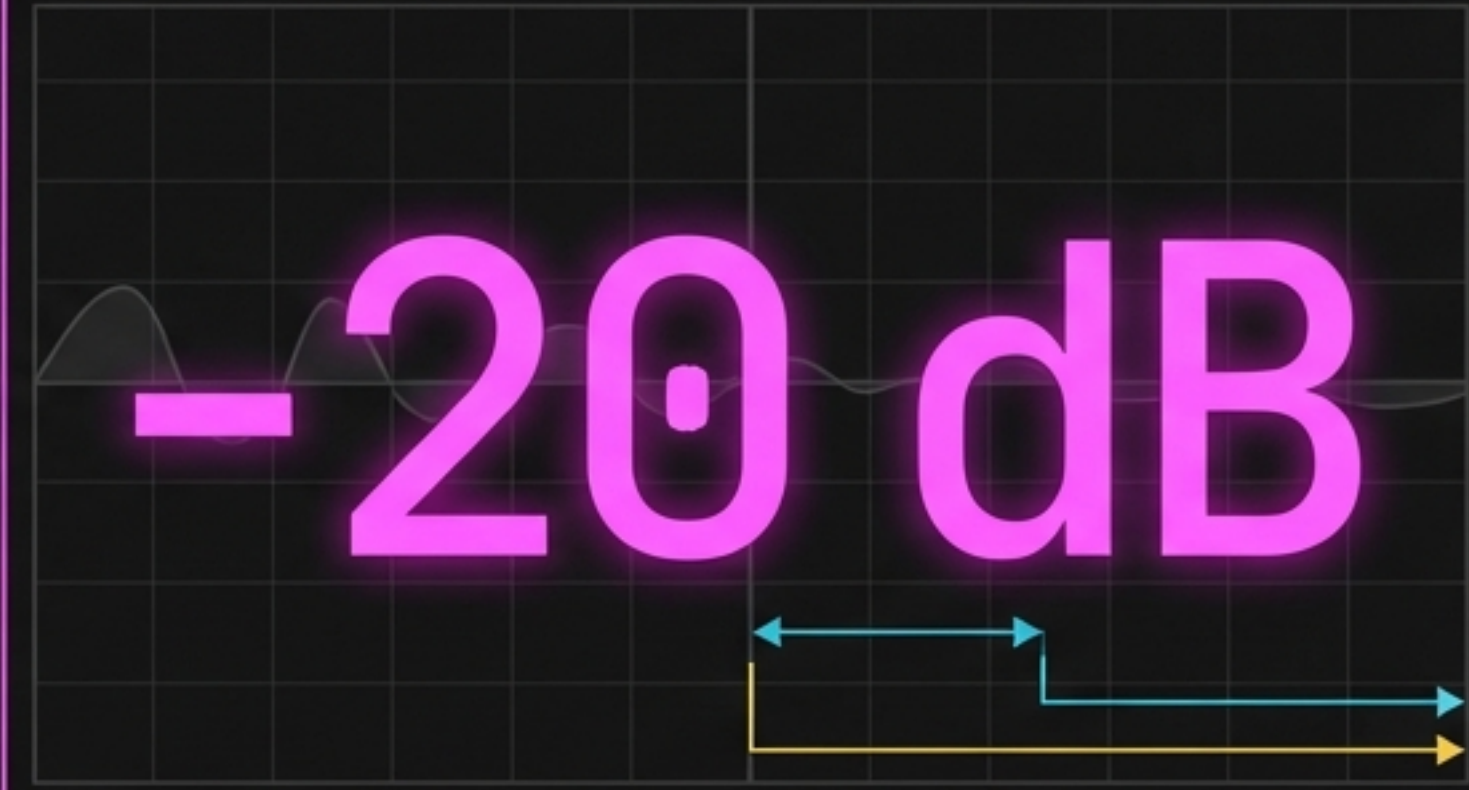
dB is RELATIVE. It measures ratios, not absolute numbers.

The Golden Rules of Mixing



-6 dB

HALVING AMPLITUDE (1/2)

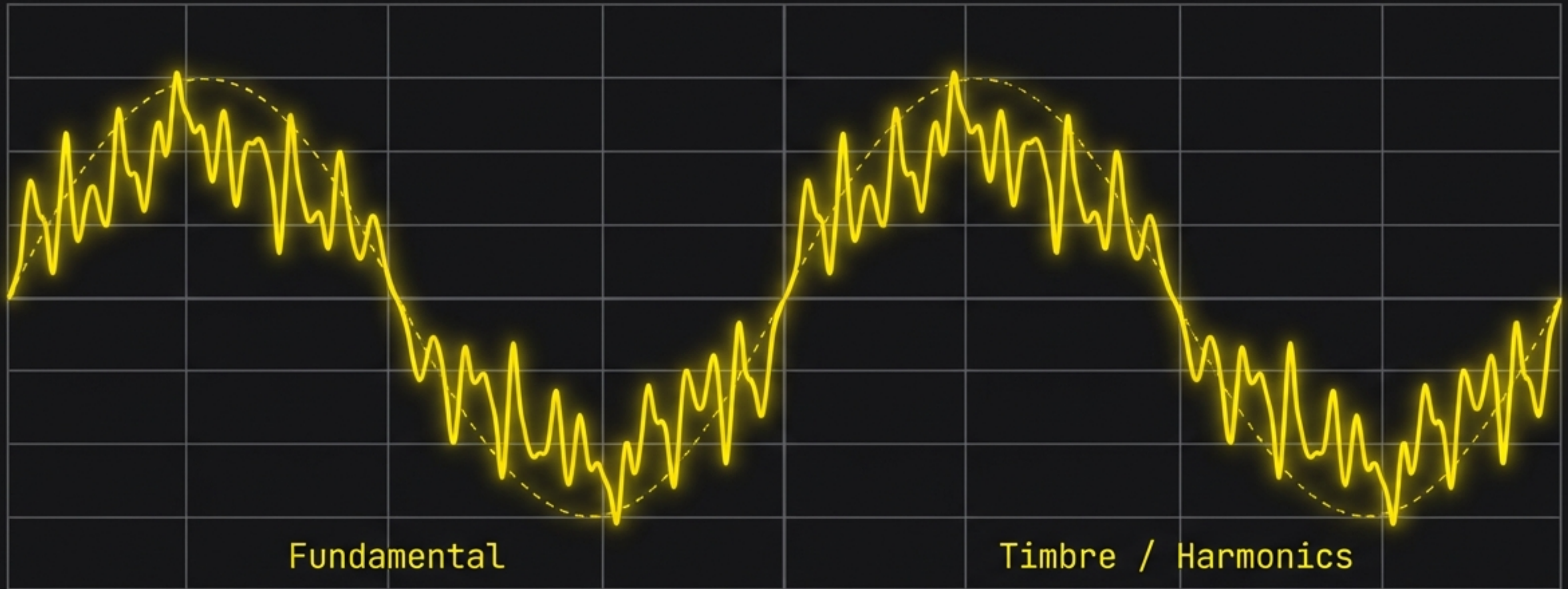


-20 dB

ONE TENTH AMPLITUDE (1/10)

Memorize these for gain staging.

Dimension 3: Shape (Timbre)



Pitch is the skeleton. Timbre is the skin.

Timbre is determined by the WAVEFORM shape.

The Four Pillars of Synthesis



SINE

Pure. No harmonics.



SQUARE

Hollow. Odd harmonics.



SAWTOOTH

Bright. All harmonics.



TRIANGLE

Mellow. Weak odd harmonics.

Breaking Symmetry: Duty Cycle



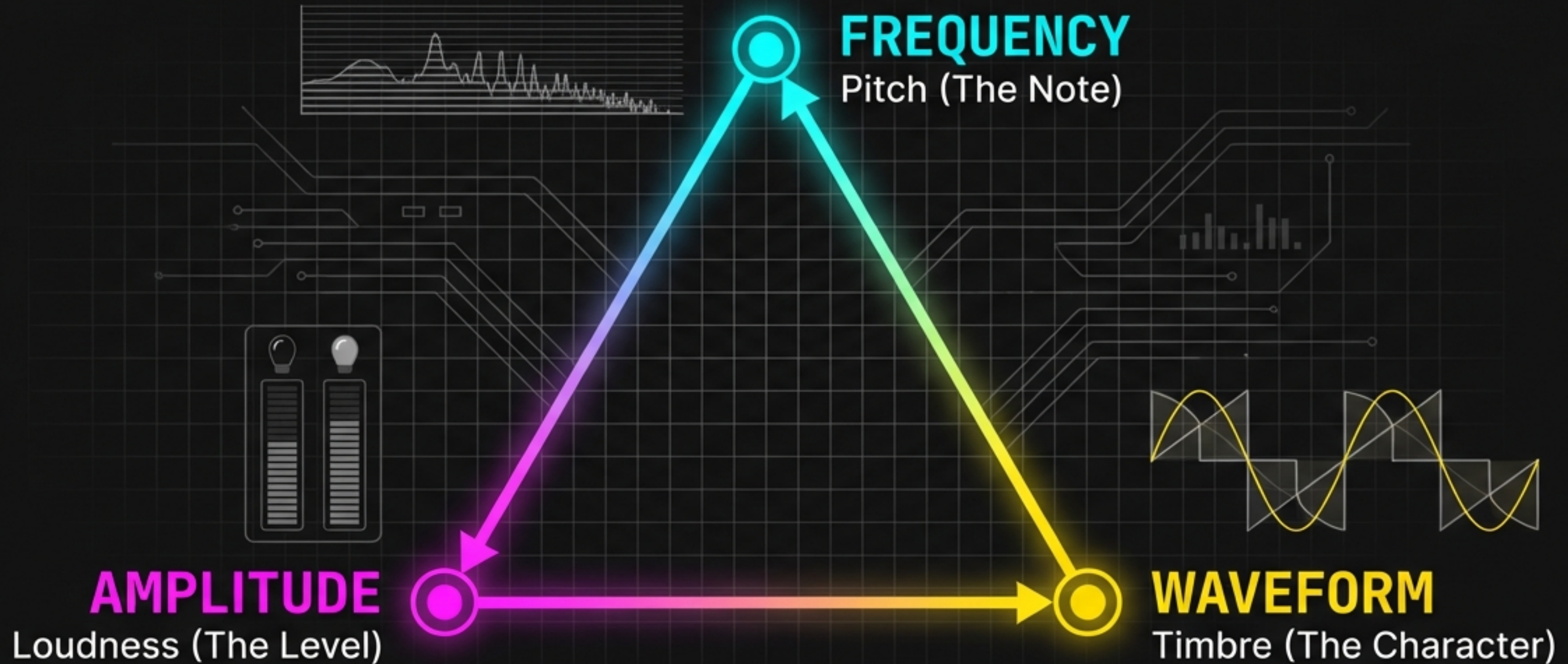
Hollow Sound



JetBrains Mono: Nasal / Thin Sound

Duty Cycle = Ratio of positive phase to negative phase.

The Producer's Triad



Every sound you produce is a manipulation of these three variables.